Islands in isogeny graphs with level structure

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Abstract

In this paper we will extend some results obtained by Love and Boneh in the paper "supersingular curves with small non-integer endomorphisms" [1] to the case of M-small supersingular elliptic curves with level-N structure. This gives a way to understand a special class of supersingular elliptic curves with level structure and gives some way to understand distance between these curves defined in terms of isogenies between them. We also present one observation that gives very simple way of describing isogenies for elliptic curves defined over the field of complex numbers.

1 Organisation

In section 3 we try to include all the preliminary information that's needed to understand the rest of the paper. Reader who is comfortable with these concepts can skip preliminary section.

In sections 4 and 5 we extend the notion of distance between supersingular curves to ones with level structure, examine some properties of this notion, and explain some methods of how to work around computations of distance.

Finally, in sections 6 and 7 we give proofs of main theorems of this paper.

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Introduction

Let N be a square-free positive integer, and consider some fixed counting of order N cyclic subgroups of $\mathbb{Z}/N\mathbb{Z} \oplus \mathbb{Z}/N\mathbb{Z}$. Let S be a set such that elements of S correspond to order N cyclic subgroups of $\mathbb{Z}/N\mathbb{Z} \oplus \mathbb{Z}/N\mathbb{Z}$. Let J denote the set of j-invariants of supersingular elliptic curves. Let E be a supersingular elliptic curve defined over $\overline{\mathbb{F}}_p$ for some prime p. Let G_i be an order N cyclic subgroup of $E[N] \cong \mathbb{Z}/N\mathbb{Z} \oplus \mathbb{Z}/N\mathbb{Z}$.

Definition 1.1 Define the distance between supersingular elliptic curves with level-N structure in the following way:

$$d((E,G),(E',G')) := \min \left\{ \deg \phi \;\middle|\; \begin{array}{l} \phi : E \to E' \text{ such that } \phi(G) = G', \text{ or} \\ \phi : E \to F_p(E') \text{ such that } \phi(G) = G'^{(p)}, \text{ or} \\ \phi : E \to D_J(E') \text{ such that } \phi(G) = D_J(G') \end{array} \right\},$$

where F_p is Frobenius involution and D_J is involution defined in [2, Chapter 4].

Definition 1.2 Define

$$S_{M,N} := \{(j,s) \in J \times S \,|\, \exists \alpha \in \mathrm{End}(E_j) \setminus \mathbb{Z},\ \alpha(G_s) \subseteq G_s,\ \deg(\alpha) \leq M\}\,.$$

The following is motivated by main theorem from[1].

2 Main results

Theorem A Suppose $pN > 4M^2$, and let $S_{M,N}$ be defined as above.
Then there exists a partition

$$S_{M,N} = \bigsqcup_D T_D$$

of $S_{M,N}$ into nonempty subsets, indexed by fundamental discriminants $-4M \leq D < 0$ which are not congruent to a square mod p. This partition has the following properties:

(a) If $(j,s), (j',s')$ are in distinct subsets $T_D \neq T_{D'}$, then (definition 4.3)

$$d((j,s),(j',s')) \geq \frac{\sqrt{pN}}{2M}.$$

(b) If $(j,s), (j',s')$ are in the same subset T_D, then there is a chain

$$(j,s) = (j_0,s_0), (j_1,s_1) \ldots, (j_r,s_r) = (j',s')$$

of elements of T_D such that

$$d((j_{i-1},s_{i-1}),(j_i,s_i)) \leq \frac{4}{\pi}\sqrt{M}$$

for all $i = 1, \ldots, r$. We can find such a chain with $r \leq 3$. Moreover, we also have

$$d((j,s),(j',s')) \leq \frac{8}{\pi}M.$$

Additionally, we have that there exists some index i such that

$$d((j_{i-1},s_{i-1}),(j_i,s_i)) \leq \sqrt[3]{\frac{8M}{\pi}}.$$

Theorem B Let E be an elliptic curve defined over $\mathbb{C}$, and let

$$\varphi : E \longrightarrow E$$

be an isogeny. Then there exists a point $P \in E(\mathbb{C})$ such that the knowledge of P and $\varphi(P)$ uniquely determines the isogeny φ. In other words, an isogeny $\varphi : E(\mathbb{C}) \to E(\mathbb{C})$ can be completely described by two pairs of coordinates.

3 Preliminaries

3.1 Elliptic curves over $\mathbb{C}$

Definition 3.1.1(Elliptic Curve over $\mathbb{C}$) An *elliptic curve* over $\mathbb{C}$ is a nonsingular curve given by a Weierstrass equation

$$E: \quad y^2 = x^3 + Ax + B,$$

where $A, B \in \mathbb{C}$ and

$$4A^3 + 27B^2 \neq 0.$$

The point at infinity

$$\mathcal{O} = [0:1:0]$$

is taken as the identity element of the elliptic curve.

Definition 3.1.2 (Fundamental Parallelogram) Let $\Lambda \subset \mathbb{C}$ be a lattice generated by two $\mathbb{R}$-linearly independent complex numbers ω_1, ω_2:

$$\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2.$$

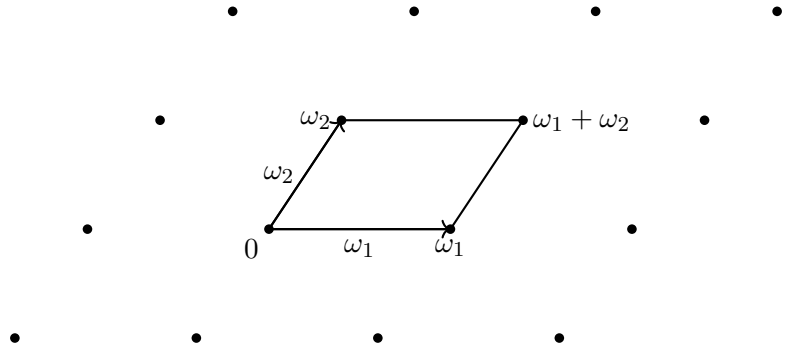
A *fundamental parallelogram* for Λ is the set

$$\mathcal{P} = \{s\omega_1 + t\omega_2 \mid 0 \leq s < 1, \quad 0 \leq t < 1\}.$$

Every point $z \in \mathbb{C}$ can be written uniquely as

$$z = p + \lambda, \qquad p \in \mathcal{P}, \quad \lambda \in \Lambda.$$

Thus, $\mathcal{P}$ contains exactly one representative from each equivalence class of $\mathbb{C}/\Lambda$.



Definition 3.1.3 (Weierstrass $\wp$-Function) Let

$$\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2 \subset \mathbb{C}$$

be a lattice. The *Weierstrass $\wp$-function* associated to Λ is defined by

$$\wp(z) = \frac{1}{z^2} + \sum_{\substack{\omega \in \Lambda \\ \omega \neq 0}} \left(\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} \right).$$

It is a meromorphic, Λ-periodic function on $\mathbb{C}$ with double poles precisely at the points of Λ. The Weierstrass $\wp$-function satisfies the differential equation

$$\big(\wp'(z)\big)^2 = 4\wp(z)^3 - g_2\wp(z) - g_3,$$

where the invariants g_2 and g_3 are given by

$$g_2 = 60 \sum_{\substack{\omega \in \Lambda \\ \omega \neq 0}} \frac{1}{\omega^4}, \qquad g_3 = 140 \sum_{\substack{\omega \in \Lambda \\ \omega \neq 0}} \frac{1}{\omega^6}.$$

Definition 3.1.4 The Weierstrass $\wp$-function induces a map

$$\Phi : \mathbb{C}/\Lambda \longrightarrow E(\mathbb{C}),$$

where

$$E : \quad y^2 = 4x^3 - g_2 x - g_3.$$

Explicitly,

$$\Phi(z) = \begin{cases} [\wp(z) : \wp'(z) : 1], & z \not\equiv 0 \pmod{\Lambda}, \\ [0 : 1 : 0], & z \equiv 0 \pmod{\Lambda}. \end{cases}$$

Thus, the point $0 \in \mathbb{C}/\Lambda$ is mapped to the point at infinity

$$\mathcal{O} = [0 : 1 : 0].$$

The map Φ is an isomorphism, hence

$$\mathbb{C}/\Lambda \cong E(\mathbb{C}).$$

This will be crucial ingredient for proving Theorem B. For more details, check [5].

3.2 Quaternion algebras

Definition 3.2.1 [4, Chapter 2.2]
An algebra B over F is a *quaternion algebra* if there exist $i, j \in B$ such that

$$1, i, j, ij$$

is an F-basis for B, and

$$i^2 = a, \qquad j^2 = b, \qquad ji = -ij$$

for some $a, b \in F^\times$.

We will primarily be interested in the case where $F = \mathbb{Q}$. For a quaternion algebra B over F, there are two natural functions that we will consider: the *reduced trace* and the *reduced norm*.

Definition 3.2.2 (Reduced Trace) Let

$$B = F + Fi + Fj + Fij$$

be a quaternion algebra over F, where

$$i^2 = a, \qquad j^2 = b, \qquad ji = -ij.$$

For an element

$$\alpha = x + yi + zj + wij \in B,$$

the *reduced trace* of α is defined by

$$\mathrm{Trd}(\alpha) = 2x.$$

Equivalently, if

$$\overline{\alpha} = x - yi - zj - wij$$

denotes the standard quaternionic conjugate of α, then

$$\mathrm{Trd}(\alpha) = \alpha + \overline{\alpha}.$$

Definition 3.2.3 (Reduced Norm) Let $\alpha = x + yi + zj + wij \in B$. The *reduced norm* of α is defined by

$$\mathrm{Nrd}(\alpha) = \alpha\overline{\alpha}.$$

Explicitly,

$$\mathrm{Nrd}(\alpha) = x^2 - ay^2 - bz^2 + abw^2.$$

We will primarily be interested in the case where $F = \mathbb{Q}$, the field of rational numbers.

3.3 Supersingular Elliptic curves with level-N structure

Definition 3.3.1 (Supersingular Elliptic Curve) Let E be an elliptic curve over $\overline{\mathbb{F}}_p$. We say that E is *supersingular* if

$$E[p] = \{Q \in E \mid pQ = \mathcal{O}\} = \{0\},$$

where $\mathcal{O}$ denotes the additive identity of the elliptic curve.
Definition 3.3.2 (Supersingular Elliptic Curve with Level-N Structure) Let E be a supersingular elliptic curve over $\overline{\mathbb{F}}_p$. Supersingular elliptic curve with level-N structure is a pair

$$(E, G),$$

where $E/\overline{\mathbb{F}}_p$ is a supersingular elliptic curve and $G \subseteq E[N]$ is a cyclic subgroup of order N.
Definition 3.3.3 [2] Let p be a prime and let N be an integer such that $\gcd(N, p) = 1$. We define $|\mathcal{S}_N|$ to be the set of equivalence classes of pairs (E, G), where E is a supersingular elliptic curve over $\overline{\mathbb{F}}_p$. Two pairs (E_1, G_1) and (E_2, G_2) are considered equivalent under the relation $\sim$ if there exists an isomorphism

$$\rho : E_1 \longrightarrow E_2$$

such that

$$\rho(G_1) = G_2.$$

The elements of $|\mathcal{S}_N|$ are called *supersingular elliptic curves with level-N structure.*
Definition 3.3.4 (Ring of Endomorphisms of a Pair) Let $(E, G) \in |\mathcal{S}_N|$. As a subring of $\mathrm{End}(E)$, we define the ring of endomorphisms of the pair (E, G) by

$$\mathcal{O}(E, G) := \{\alpha \in \mathrm{End}(E) : \alpha(G) \subseteq G\}.$$

In other words, $\mathcal{O}(E, G)$ consists of those endomorphisms of E which respect the level-N structure G.

Throughout this paper, we restrict N to be square-free and coprime with p, where E is defined over $\overline{\mathbb{F}}_p$.

3.4 Deuring correspondence (for supersingular elliptic curves with level-N structure)

Definition 3.4.1 (Deuring Correspondence [2]) Let $E/\overline{\mathbb{F}}_p$ be a supersingular elliptic curve. The *Deuring correspondence* associates to E the maximal order

$$\mathrm{End}(E)$$

in the quaternion algebra

$$B_{p,\infty},$$

the unique quaternion algebra over $\mathbb{Q}$ ramified exactly at p and at infinity.

More generally, the Deuring correspondence gives a bijection between supersingular elliptic curves over $\overline{\mathbb{F}}_p$ up to isomorphism and maximal orders in $B_{p,\infty}$ up to conjugation.

Definition 3.4.2 (Deuring Correspondence with Level-N Structure [2]) Let $E/\overline{\mathbb{F}}_p$ be a supersingular elliptic curve equipped with a level-N structure $G \subseteq E[N]$. The *Deuring correspondence with level-N structure* associates to the pair

$$(E, G)$$

an Eichler order

$$\mathcal{O}(E, G) \subseteq \mathrm{End}(E)$$

of level N in the quaternion algebra $B_{p,\infty}$.

More precisely, the correspondence gives a bijection between isomorphism classes of supersingular elliptic curves with level-N structure and conjugacy classes of Eichler orders of level N in $B_{p,\infty}$. The maximal order

$$\mathrm{End}(E)$$

corresponds to the supersingular elliptic curve E, while the additional level structure $G \subseteq E[N]$ determines the choice of the Eichler suborder.

One proposition which allows us to understand why this correspondence occurs is the following:

Proposition 3.4.3 (Proposition 3.4, [2]) Let $(E, G) \in |\mathcal{S}_N|$ and let

$$\phi : E \longrightarrow E/G$$

be an isogeny with

$$\ker(\phi) = G.$$

Then

$$\mathcal{O}(E, G) = \mathrm{End}(E) \cap \left(\frac{1}{\deg(\phi)} \phi \, \mathrm{End}(E/G) \, \widehat{\varphi} \right),$$

where the intersection is taken inside the quaternion algebra $B_{p,\infty}$ via the embedding described in [2](2.3).

4 Distance between Eichler orders

We follow paper by Love and Boneh and make adjustments where needed.

Definition 4.1 An Eichler order $\mathcal{O} \subset B$ is called *M-small* if there exists an element

$$\alpha \in \mathcal{O} \setminus \mathbb{Z}$$

such that

$$\mathrm{Nrd}(\alpha) \leq M.$$

Lemma 4.2 Let $\mathcal{N}, \mathcal{N}' \subset B$ be Eichler orders of same level N. Then the following quantities are all equal:

(a) $|\mathcal{N} : \mathcal{N} \cap \mathcal{N}'|$, the index of $\mathcal{N} \cap \mathcal{N}'$ in $\mathcal{N}$;

(b) $|\mathcal{N}' : \mathcal{N} \cap \mathcal{N}'|$, the index of $\mathcal{N} \cap \mathcal{N}'$ in $\mathcal{N}'$;

(c) The smallest reduced norm of an integral ideal linking $\mathcal{N}$ to $\mathcal{N}'$.

Proof: We note that these quantities are equal if and only if they are equal at each prime localization. First, we note that if we localize at a prime p, where the quaternion algebra B is ramified, then we get that if $\mathcal{N}$ inside B is an Eichler order, then $\mathcal{N}_p$ inside B_p has to be an Eichler order, but since B ramifies at p there exists only 1 maximal order in B_p, hence an Eichler order can only be this unique maximal order and from here it's clear that the quantities above are equal.

Now, let us consider the case where $\ell \neq p$ is a prime number where localization happens. Then by ([4] 23.4) we get that $\mathcal{N}_\ell$ is isomorphic to

$$\begin{pmatrix} \mathbb{Z}_\ell & \mathbb{Z}_\ell \\ \ell^e\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix}$$

where e is the power of ℓ that divides N, so it's 1 or 0, since N is square-free. $\mathcal{N}'_\ell$ is isomorphic to

$$\alpha \begin{pmatrix} \mathbb{Z}_\ell & \mathbb{Z}_\ell \\ \ell^e\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix} \alpha^{-1},$$

for some α, which, without loss of generality(by scaling), is assumed to belong to

$$\begin{pmatrix} \mathbb{Z}_\ell & \mathbb{Z}_\ell \\ \mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix} \setminus \begin{pmatrix} \ell\mathbb{Z}_\ell & \ell\mathbb{Z}_\ell \\ \ell\mathbb{Z}_\ell & \ell\mathbb{Z}_\ell \end{pmatrix}$$

There exist (Smith normal form) $\beta', \gamma' \in \mathrm{GL}_2(\mathbb{Z}_\ell)$ [4, 23.4.3] such that

$$\beta'\alpha\gamma' = \begin{pmatrix} 1 & 0 \\ 0 & \ell^e \end{pmatrix}.$$

Setting

$$\beta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \beta' \qquad \text{and} \qquad \gamma = \gamma' \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

we find that

$$\beta\alpha\gamma = \begin{pmatrix} \ell^e & 0 \\ 0 & 1 \end{pmatrix}.$$

Then

$$\mathcal{N}_\ell \cap \mathcal{N}'_\ell \simeq \mathcal{N}_\ell \cap \beta\mathcal{N}'_\ell\beta^{-1} = \begin{pmatrix} \mathbb{Z}_\ell & \mathbb{Z}_\ell \\ \ell^e\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix} \cap \beta\alpha \begin{pmatrix} \mathbb{Z}_\ell & \mathbb{Z}_\ell \\ \ell^e\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix} \alpha^{-1}\beta^{-1}.$$

Now, looking at the second term of the intersection, we note that

$$\beta\alpha\gamma \begin{pmatrix} \mathbb{Z}_\ell & \mathbb{Z}_\ell \\ \ell^e\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix} \gamma^{-1}\alpha^{-1}\beta^{-1} = \begin{pmatrix} \ell^e & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \mathbb{Z}_\ell & \mathbb{Z}_\ell \\ \ell^e\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix} \begin{pmatrix} \ell^{-e} & 0 \\ 0 & 1 \end{pmatrix}.$$

$$= \begin{pmatrix} \mathbb{Z}_\ell & \ell^e\mathbb{Z}_\ell \\ \mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix}.$$

Then

$$\mathcal{N}_\ell \cap \mathcal{N}'_\ell = \begin{pmatrix} \mathbb{Z}_\ell & \mathbb{Z}_\ell \\ \ell^e\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix} \cap \begin{pmatrix} \mathbb{Z}_\ell & \ell^e\mathbb{Z}_\ell \\ \mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix}. = \begin{pmatrix} \mathbb{Z}_\ell & \ell^e\mathbb{Z}_\ell \\ \ell^e\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix}.$$

And finally, the index calculation gives:

$$\left|\mathcal{N}_\ell : \mathcal{N}_\ell \cap \mathcal{N}'_\ell\right| = \left| \begin{pmatrix} \mathbb{Z}_\ell & \mathbb{Z}_\ell \\ \ell^e\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix} : \begin{pmatrix} \mathbb{Z}_\ell & \ell^e\mathbb{Z}_\ell \\ \ell^e\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix}. \right| = l^e.$$

By symmetry of the proof we will have that

$$\left|\mathcal{N}_\ell : \mathcal{N}_\ell \cap \mathcal{N}'_\ell\right| = \left|\mathcal{N}'_\ell : \mathcal{N}_\ell \cap \mathcal{N}'_\ell\right|.$$

Now we need to understand the smallest reduced norm of an integral linking ideal, linking $\mathcal{N}_\ell$ to $\mathcal{N}'_\ell$. We can construct such an ideal I at each localization and finally take the intersection over all localizations, and ultimately construct I. Since $\mathcal{N}'_\ell = w\mathcal{N}_\ell w^{-1}$ for $w = \begin{pmatrix} \ell^e & 0 \\ 0 & 1 \end{pmatrix}$, set

$$I_\ell = \mathcal{N}'_\ell w \cap \mathcal{N}_\ell = \begin{pmatrix} \ell^e\mathbb{Z}_\ell & \ell^e\mathbb{Z}_\ell \\ \ell^e\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix} \cap \begin{pmatrix} \mathbb{Z}_\ell & \mathbb{Z}_\ell \\ \ell^e\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix} = \begin{pmatrix} \ell^e\mathbb{Z}_\ell & \ell^e\mathbb{Z}_\ell \\ \ell^e\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix},$$

which has smallest reduced norm ℓ^e, realized at the matrix

$$\begin{pmatrix} \ell^e & 0 \\ 0 & 1 \end{pmatrix}.$$

Finally, setting

$$I = \bigcap_q I_q,$$

where the intersection is taken over all primes q gives the desired linking integral ideal from $\mathcal{N}$ to $\mathcal{N}'$.

Definition 4.3 Let $\mathcal{N}, \mathcal{N}' \subset B$ be Eichler orders. We define the *distance* between $\mathcal{N}$ and $\mathcal{N}'$ by

$$d(\mathcal{N}, \mathcal{N}') = \left|\mathcal{N} : \mathcal{N} \cap \mathcal{N}'\right|.$$

Lemma 4.4 Let (E, G) and (E', G') be supersingular elliptic curves with level-N structure. Then

$$d((E,G),(E',G')) = \min\left\{d(\mathcal{N},\mathcal{N}') \;\middle|\; \mathcal{N} \cong \mathcal{O}(E,G),\ \mathcal{N}' \cong \mathcal{O}(E',G')\right\}.$$

Proof. We claim that both of these quantities are equal to

$$\min\left\{\deg(\varphi) \;\middle|\; \begin{array}{l} \varphi : E \longrightarrow E'' \text{ is an isogeny for some } E'', \\ \varphi(G) = G'', \text{ and } \mathcal{O}(E',G') \cong \mathcal{O}(E'',G'') \end{array}\right\}.$$

Let's rewrite the sets:

$$A_1 = \left\{\deg\phi \;\middle|\; \begin{array}{l} \phi : E \to E' \text{ with } \phi(G) = G', \text{ or} \\ \phi : E \to E'^{(p)} \text{ with } \phi(G) = G'^{(p)}, \text{ or} \\ \phi : E \to D_J(E') \text{ with } \phi(G) = D_J(G') \end{array}\right\}$$

$$A_2 = \left\{\deg\phi \;\middle|\; \phi : E \to E'' \text{ such that } \phi(G) = G'' \text{ and } \mathcal{O}(E'',G'') \cong \mathcal{O}(E',G')\right\}$$

$$A_3 = \left\{d(\mathcal{N},\mathcal{N}') \;\middle|\; \mathcal{N} \cong \mathcal{O}(E,G) \text{ and } \mathcal{N}' \cong \mathcal{O}(E',G)\right\}$$

Let's prove that $A_3 = A_2$.

Take $\deg\phi \in A_2$, then there exists an integral linking ideal $I_{\phi,G}$ which links $\mathcal{O}(E,G)$ to $\mathcal{O}(E'',G'') \cong \mathcal{O}(E',G')$ (viewed as Eichler orders in quaternion algebra) and satisfies $\mathrm{nrd}(I_{\phi,G}) = \deg(\phi)$. As we proved earlier, the smallest reduced norm of an integral ideal linking two Eichler orders $\mathcal{N}$ and $\mathcal{N}'$ is the distance between them, and because of this we find that $\deg\phi$ belongs to A_3.

Now take an element from A_3, call it r, which is realized at $\mathcal{N}$ and $\mathcal{N}'$ for some $\mathcal{N} \cong \mathcal{O}(E,G)$ and $\mathcal{N}' \cong \mathcal{O}(E',G')$. Write

$$\mathcal{N} = \alpha\mathcal{O}(E,G)\alpha^{-1}$$

and

$$\mathcal{N}' = \beta\mathcal{O}(E',G')\beta^{-1}.$$

Then

$$r = d\left(\alpha\mathcal{O}(E,G)\alpha^{-1}, \beta\mathcal{O}(E',G')\beta^{-1}\right) = d\left(\mathcal{O}(E,G), \alpha^{-1}\beta\mathcal{O}(E',G')\beta^{-1}\alpha\right) =$$

= smallest reduced norm of an integral ideal connecting these two Eichler orders = $\deg \phi$ for some $\phi : E \to E'$ with $\phi(G) = G'$, and this gives that r belongs to A_2, hence $A_2 = A_3$.

Let's now prove that $A_1 = A_2$.

Take $r \in A_1$. This will correspond to some isogeny $\phi : E \to E''$ where $\mathcal{O}(E'', G'') \cong \mathcal{O}(E', G')$ by [2, Theorem 4.13].

Now take $r \in A_2$. This also corresponds to some isogeny $\phi : E \to E''$ with $\phi(G) = G''$ and $\mathcal{O}(E'', G'') \cong \mathcal{O}(E', G')$. Since all the preimages of the map $\mathcal{O}(\cdot, \cdot)$ lie in the definition of A_1 (namely, $\mathcal{O}(E', G') = \mathcal{O}(E'', G'')$ implies that $(E'', G'') \cong (E', G')$ or $D_J(E', G')$ or $F_p(E', G')$), we conclude that r belongs to A_1 and hence $A_1 = A_2$.

Given that $A_1 = A_2 = A_3$, we find that taking minima on all of these sets will be equal, and this completes the proof.

□

Claim. Suppose $\mathcal{N}$ and $\mathcal{N}'$ are both M-small Eichler orders. Let

$$\alpha \in \mathcal{N} \setminus \mathbb{Z} \qquad \text{and} \qquad \alpha' \in \mathcal{N}' \setminus \mathbb{Z}$$

be elements satisfying

$$\mathrm{Nrd}(\alpha) \leq M \qquad \text{and} \qquad \mathrm{Nrd}(\alpha') \leq M.$$

We will show that the distance between $\mathcal{N}$ and $\mathcal{N}'$ is small if

$$\mathbb{Q}(\alpha) \cong \mathbb{Q}(\alpha'),$$

and large otherwise.

Following is modified version of Theorem 2 from [3].

Theorem 4.5 Let $\mathcal{N} \subset B$ be Eichler order of square-free level. Suppose that $\mathcal{O}$ and $\mathcal{O}'$ are quadratic orders in imaginary quadratic fields, optimally embedded into $\mathcal{N}$ with distinct images. Then

$$\mathrm{disc}(\mathcal{O}) \, \mathrm{disc}(\mathcal{O}') \geq 4pN.$$

Proof. We follow the proof of theorem 2 from [3], using the fact that embeddings preserve the trace and norm of elements.

Suppose that the quadratic orders $\mathcal{O}_1$ and $\mathcal{O}_2$ are optimally embedded into Eichler order $\mathcal{N}$, with distinct images. As in the original proof, we consider the elements

$$1, \alpha_1, \alpha_2, \alpha_1\alpha_2$$

where α_i is the image of $\frac{1}{2}(D_i + \sqrt{-D_i})$ after embedding $\mathcal{O}$ and $\mathcal{O}'$ into $\mathcal{N}$, with D_i being the discriminant of the corresponding quadratic order. We consider the $\mathbb{Z}$-module generated by them. The computation of the discriminant of this module is unchanged by the fact that the quadratic orders are embedded into Eichler orders rather than maximal orders. Indeed, the computation depends only on the traces and norms of the elements, which are preserved under the embeddings.

Thus, we obtain the same expression for the discriminant of the order generated by these elements as in Kaneko's proof. Since the discriminant of an order contained in an Eichler order of level N in the quaternion algebra $B_{p,\infty}$ must be divisible by $(pN)^2$, we obtain

$$pN \mid \frac{1}{4}\left(\mathrm{disc}(\mathcal{O}) \, \mathrm{disc}(\mathcal{O}') - \left(2s - \mathrm{disc}(\mathcal{O}) \, \mathrm{disc}(\mathcal{O}')\right)^2\right),$$

where

$$s = \mathrm{Trd}(\alpha_1\alpha_2).$$

is an integer. In particular,

$$pN \leq \frac{1}{4}\left(\mathrm{disc}(\mathcal{O}) \, \mathrm{disc}(\mathcal{O}') - \left(2s - \mathrm{disc}(\mathcal{O}) \, \mathrm{disc}(\mathcal{O}')\right)^2\right) \leq \frac{1}{4} \mathrm{disc}(\mathcal{O}) \, \mathrm{disc}(\mathcal{O}').$$

This gives the desired lower bound.

□

Proof. We now return to the proof of the case when $\mathbb{Q}(\alpha) \ncong \mathbb{Q}(\alpha')$. In this case, if we follow the proof of proposition 4.5 by Love and Boneh [1], we find that

$$d^2 \geq \frac{\operatorname{disc}(\mathcal{O})}{4\operatorname{Nrd}(\alpha)} \frac{\operatorname{disc}(\mathcal{O}')}{4\operatorname{Nrd}(\alpha')} \geq \frac{pN}{4M^2}.$$

□

So, the distance between N and N' is large in this case. Proof of second statement will be given in the end of section 5. Before that we need to understand steps.

5 Steps

In this section we will extend how to step between different Eichler orders which admit same and different embeddings of quadratic orders. We follow section 5 in paper by Love and Boneh.

First, we consider the case in which the same quadratic order $\mathcal{O}$ is optimally embedded in two Eichler orders $\mathcal{N}$ and $\mathcal{N}'$.

Theorem 5.1(Theorem 5.2 [7]) Let $\mathcal{N}, \mathcal{N}' \subset B$ be two Eichler orders of the same square-free level, and suppose

$$\mathcal{O} \cong K \cap \mathcal{N} = K \cap \mathcal{N}'$$

is optimally embedded in each. Then there exists an invertible ideal $\mathfrak{a}$ of $\mathcal{O}$ such that

$$\mathcal{N}\mathfrak{a} = \mathfrak{a}\mathcal{N}'.$$

In the paper by Love and Boneh, a weaker version of this result is used, namely the case in which $\mathcal{N}, \mathcal{N}'$ are maximal orders.

Theorem 5.2 [Skolem–Noether, [4]] Let A and B be simple F-algebras, and suppose that B is central over F. If

$$f, g : A \longrightarrow B$$

are F-algebra homomorphisms, then there exists an element $\beta \in B^\times$ such that

$$f(\alpha) = \beta^{-1} g(\alpha) \beta$$

for all $\alpha \in A$.

We can follow Lemma 5.3 from the paper of Love and Boneh [1], noting that the proof generalizes naturally to Eichler orders of the same square-free level.

By the Skolem–Noether theorem, there exists $\gamma \in B^\times$ such that

$$\mathcal{N} \cap K = (\gamma^{-1}\mathcal{N}'\gamma) \cap K.$$

Moreover, Theorem 5.2 guarantees the existence of an invertible ideal $\mathfrak{a}$ of $\mathcal{O}$ such that

$$\mathcal{N}\mathfrak{a} = \mathfrak{a}\mathcal{N}'.$$

All other steps of the proof of Lemma 5.3 [1] can then be repeated with Eichler orders in place of maximal orders. Consequently, there exists

$$\varphi : B \longrightarrow B$$

such that

$$d\big(\mathcal{N}, \varphi(\mathcal{N}')\big) \leq \frac{2}{\pi} f \sqrt{|D|}.$$

where D is discriminant of quadratic order $\mathcal{O}$ and f is conductor of the quadratic order

$$\mathcal{O} = \mathcal{O}_{K,f}$$

which optimally embeds into same square-free level Eichler orders $\mathcal{N}$ and $\mathcal{N}'$. Then the distance between them satisfies

$$d(\mathcal{N}, \mathcal{N}') \leq \frac{2}{\pi} f \sqrt{|D|}.$$

This will be referred to as a horizontal step between $\mathcal{N}$ and $\mathcal{N}'$.

Definition 5.3 (Vertical Steps) Again following the paper by Love and Boneh, we need to determine how to step between Eichler orders which have different quadratic orders embedded in them. Following lemma is modification of lemma 5.4 [1].

Lemma 5.4 Let ℓ be a prime greater than 3, and $\beta \in \mathcal{O}_K$. Let $\mathcal{N} \subseteq B$ be an Eichler order in which $\mathbb{Z}[\ell\beta]$ optimally embeds. Then there exists an Eichler order $\mathcal{N}'$ in which $\mathbb{Z}[\beta]$ optimally embeds, with $d(\mathcal{N}, \mathcal{N}') = \ell$.

Proof. Proof is a slight modification of proof of lemma 5.4 from [1].

Let us consider completions of $\mathcal{N}$ at different primes $\ell > 3$. If $\ell = p$, then we have no embedding of $\mathbb{Z}[p\beta]$ in maximal orders, hence no embedding in Eichler orders. Suppose $\ell \neq p$, then $\mathcal{N}_\ell$ inside B_ℓ is isomorphic [4][23.4.3] to

$$\begin{pmatrix} \mathbb{Z}_\ell & \mathbb{Z}_\ell \\ \ell\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix}.$$

Consider embedding of $\mathbb{Z}[\ell\beta]$ into $\begin{pmatrix} \mathbb{Z}_\ell & \mathbb{Z}_\ell \\ \ell\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix}$ given by

$$\ell\beta \mapsto \begin{pmatrix} 0 & 1 \\ -\ell^2 n & \ell t \end{pmatrix}$$

where $n = \mathrm{N}(\beta)$ and $t = \mathrm{Tr}(\beta)$. This describes completely how to embed

$$\mathbb{Z}[\ell\beta] = \left\{ a_0 + a_1(\ell\beta) + a_2(\ell\beta)^2 + \cdots \mid a_i \in \mathbb{Z} \right\}$$

into $\begin{pmatrix} \mathbb{Z}_\ell & \mathbb{Z}_\ell \\ \ell\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix}$. Note that β will get mapped to

$$\beta \mapsto \begin{pmatrix} 0 & 1/\ell \\ -\ell n & t \end{pmatrix} \subseteq \begin{pmatrix} \mathbb{Z}_\ell & \frac{1}{\ell}\mathbb{Z}_\ell \\ \ell\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix}$$

So this embedding gives a natural way to embed

$$\mathbb{Z}[\beta] = \left\{ a_0 + a_1\beta + a_2\beta^2 + \cdots \mid a_i \in \mathbb{Z} \right\}$$

into

$$\begin{pmatrix} \mathbb{Z}_\ell & \frac{1}{\ell}\mathbb{Z}_\ell \\ \ell\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix},$$

(Also note that

$$\begin{pmatrix} \mathbb{Z}_\ell & \frac{1}{\ell}\mathbb{Z}_\ell \\ \ell\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix} \begin{pmatrix} \mathbb{Z}_\ell & \frac{1}{\ell}\mathbb{Z}_\ell \\ \ell\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix} \subseteq \begin{pmatrix} \mathbb{Z}_\ell & \frac{1}{\ell}\mathbb{Z}_\ell \\ \ell\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix}.)$$

Which turns out to be conjugate to an Eichler order of level ℓ in the following way:

$$\begin{pmatrix} \mathbb{Z}_\ell & \frac{1}{\ell}\mathbb{Z}_\ell \\ \ell\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix} = \begin{pmatrix} 3 & 2/\ell \\ l & 1 \end{pmatrix} \begin{pmatrix} \mathbb{Z}_\ell & \mathbb{Z}_\ell \\ \ell\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix} \begin{pmatrix} 1 & -2/\ell \\ -\ell & 3 \end{pmatrix}.$$

Then if we define

$$\mathcal{N}'_\ell := \begin{pmatrix} \mathbb{Z}_\ell & \frac{1}{\ell}\mathbb{Z}_\ell \\ \ell\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix}$$

and further define

$$\mathcal{N}' := \mathcal{N}'_\ell \cap \left(\bigcap_{q \neq \ell} \mathcal{N}_q \right),$$

then we find that $\mathbb{Z}[\beta]$ embeds in $\mathcal{N}'$ and

$$d(\mathcal{N}', \mathcal{N}) = \prod_{\text{primes } q} d(\mathcal{N}'_q, \mathcal{N}_q) = d(\mathcal{N}'_\ell, \mathcal{N}_\ell) = \left| \mathcal{N}'_\ell : \mathcal{N}_\ell \cap \mathcal{N}'_\ell \right|$$

$$= \left| \begin{pmatrix} \mathbb{Z}_\ell & \frac{1}{\ell}\mathbb{Z}_\ell \\ \ell\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix} : \begin{pmatrix} \mathbb{Z}_\ell & \mathbb{Z}_\ell \\ \ell\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix} \cap \begin{pmatrix} \mathbb{Z}_\ell & \frac{1}{\ell}\mathbb{Z}_\ell \\ \ell\mathbb{Z}_\ell & \mathbb{Z}_\ell \end{pmatrix} \right| = l$$

Now, following Corollary 5.5 from the Love/Boneh paper, we analogously extend the result to Eichler orders and find that if f is a positive integer and N is an Eichler order in which $\mathcal{O}_{K,f}$ optimally embeds, then there exists an Eichler order N' in which $\mathcal{O}_K$ embeds and

$$d(N, N') \leq f.$$

Let's now prove the second part of the claim about distances between Eichler orders. Suppose $\alpha \in \mathcal{N} - \mathbb{Z}$ and $\alpha' \in \mathcal{N}' - \mathbb{Z}$, and $\mathbb{Q}(\alpha) \cong \mathbb{Q}(\alpha')$. Let

$$K \cong \mathbb{Q}(\alpha) \cong \mathbb{Q}(\alpha').$$

Since $\mathbb{Q}(\alpha)$ embeds in $\mathcal{N}$ and $\mathcal{N}'$, depending on $\mathcal{N}$ and $\mathcal{N}'$, there will exist f_1 and f_2 such that $\mathcal{O}_{K,f_1}$ embeds in $\mathcal{N}$ and $\mathcal{O}_{K,f_2}$ embeds in $\mathcal{N}'$. By Corollary 5.5, take vertical steps from $\mathcal{N}$ and $\mathcal{N}'$ to $\widetilde{\mathcal{N}}$ and $\widetilde{\mathcal{N}}'$, where $\widetilde{\mathcal{N}}$ and $\widetilde{\mathcal{N}}'$ admit optimal embeddings of $\mathcal{O}_K$ and

$$d(\mathcal{N}, \widetilde{\mathcal{N}}) \leq f_1, \qquad d(\mathcal{N}', \widetilde{\mathcal{N}}') \leq f_2.$$

Then, since $\mathcal{O}_K$ embeds into $\widetilde{\mathcal{N}}$ and $\widetilde{\mathcal{N}}'$, we can take a vertical step between $\widetilde{\mathcal{N}}$ and something isomorphic to $\widetilde{\mathcal{N}}'$ so that the distance between them is less than or equal to

$$\frac{2}{\pi} f \sqrt{|D|}$$

Finally, we will get a diagram:

$$\begin{array}{ccc} \mathcal{N} & & \mathcal{N}' \\ \downarrow & & \downarrow \\ \widetilde{\mathcal{N}} & & \widetilde{\mathcal{N}}' \\ & \searrow & \downarrow \\ & & f_i(\widetilde{\mathcal{N}}') \end{array}$$

Due to the fact that distance between Eichler orders can be associated with smallest reduced norms of linking ideals and properties of linking ideals, we find that

$$\begin{aligned} d(\mathcal{N}, \mathcal{N}') &\leq d(\mathcal{N}, \widetilde{\mathcal{N}}) \cdot d(\widetilde{\mathcal{N}}, f_i(\widetilde{\mathcal{N}}')) \cdot d(f_i(\widetilde{\mathcal{N}}'), \widetilde{\mathcal{N}}') \cdot d(\widetilde{\mathcal{N}}', \mathcal{N}') \\ &\leq f_1 \cdot \frac{2}{\pi} f \sqrt{|D|} \cdot 1 \cdot f_2 \\ &= \frac{2}{\pi} f_1 f_2 \sqrt{|D|} \overset{\text{AM-GM}}{\leq} \frac{2}{\pi} \frac{f_1^2 + f_2^2}{2} |D| \\ &= \frac{4}{\pi} \left(\frac{f_1^2 |D|}{4} + \frac{f_2^2 |D|}{4} \right) \overset{*}{\leq} \frac{4}{\pi} \left(\text{nrd}(\alpha) + \text{nrd}(\alpha') \right) \leq \frac{4}{\pi}(M + M) = \frac{8}{\pi} M. \end{aligned}$$

where $(*)$ is obtained using the inequality from the proof of Proposition 4.5 from [1], and the final inequality is obtained from noting that α and α' are both M-small.

Corollary. We get that some distance along the chain must be less than or equal to $\sqrt[3]{\frac{8}{\pi}M}$.

Remark. The bound on consecutive distances can be analogously proven to be less than or equal to $\frac{4}{\pi}\sqrt{M}$ as it's done in [1, proof of Proposition 4.6].

□

6 Proof of Theorem A

We claim that

$$S_{M,N} = \bigsqcup_D T_D,$$

where T_D is defined as

$$T_D = \left\{ (j,s) \in J \times S \;\middle|\; \exists\, \alpha \in \mathcal{O}(E_j, G_s) \setminus \mathbb{Z} \text{ such that } \deg(\alpha) \leq M \text{ and } \mathbb{Q}(\alpha) \cong \mathbb{Q}(\sqrt{D}) \right\}.$$

Proof. We follow the proof by Love and Boneh.

Let's prove that these sets are disjoint. If $(j,s) \in T_D$ and $(j,s) \in T_{D'}$, then there exist $\alpha, \alpha' \in \mathcal{O}(E_j, G_s) \setminus \mathbb{Z}$ such that they generate non-isomorphic subfields. For this reason, using the result obtained above on the lower bound of the distance between two such Eichler orders, we get

$$1 = d(\mathcal{N}, \mathcal{N}')^2 \geq \frac{pN}{4M^2},$$

which contradicts our assumption that

$$M < \frac{\sqrt{pN}}{2}.$$

Let's prove non-emptiness of these sets. Following the proof by Love and Boneh, we find that α is contained in some maximal order. Let's view maximal order as $\mathbb{Z}$-lattice of the form

$$\mathbb{Z}w_1 + \mathbb{Z}w_2 + \mathbb{Z}w_3 + \mathbb{Z}w_4,$$

where $\{w_i\}$ is a basis for maximal order.

Let's note that $\mathbb{Q}(\alpha)$, after being embedded into B will be 2 dimensional over $\mathbb{Q}$. As ring of integers of $\mathbb{Q}(\alpha)$ is 2 dimensional lattice, this means that after embedding ring of integers of $\mathbb{Q}(\alpha)$ into a maximal order, we will only use span of 2 vectors out of 4 basis vectors from $\{w_i\}$. This implies that α is of the form

$$\alpha = k_1 w_{i_1} + k_2 w_{i_2}.$$

Without loss of generality let w_1 and w_2 be used basis vectors (equivalently, choose a basis so that $\mathbb{Q}(\alpha)$ lies in the $\mathbb{Q}$-span of w_1 and w_2.). Now it is easy to notice that α will still belong to the lattice given by

$$L' = \mathbb{Z}w_1 + \mathbb{Z}w_2 + \mathbb{Z}w_3 + \mathbb{Z}Nw_4.$$

Let's now show that L' is an Eichler order of level N.

Localising at each prime we get that L'_r is isomorphic to L_r when r does not divide N, so

$$d(L'_r, L_r) = 1.$$

Consider primes dividing N. If r divides N, then localising at r gives that

$$L'_r \cong \mathbb{Z}w_1 + \mathbb{Z}w_2 + \mathbb{Z}w_3 + \mathbb{Z}rw_4 \subseteq \mathbb{Z}w_1 + \mathbb{Z}w_2 + \mathbb{Z}w_3 + \mathbb{Z}w_4.$$

By the theory of Bruhat–Tits tree [4][23.5] we find that L_r and L'_r are neighbours and this lets us conclude that

$$d(L_r, L'_r) = r.$$

Since

$$d(L, L') = \prod_{r\text{-prime}} d(L_r, L'_r),$$

this lets us conclude that

$$d(L, L') = N,$$

so L' is Eichler order and contains α.
We then make an embedding

$$\iota : \mathbb{Z}[\alpha] \hookrightarrow \mathcal{O}(E_j, G_s)$$

for some (j, s) and since

$$\mathbb{Q}(\iota(\alpha)) \cong \mathbb{Q}(\sqrt{D}),$$

we get that there exists some $(j, s) \in T_D$, so the sets T_D are non-empty.

Let's finally prove that $S_{M,N} \subseteq \bigsqcup_D T_D$. Take $(j, s) \in S_{M,N}$. Then there exists $\alpha \in \mathrm{End}(E_j) - \mathbb{Z}$, such that $\deg(\alpha) \leq M$ and $\alpha(G_s) \subseteq G_s$. Then, following the proof by Love and Boneh, and using first two properties of α, we find that $\mathbb{Q}(\alpha) \cong \mathbb{Q}(\sqrt{D})$ for some discriminant D, and this together with the fact that $\alpha(G_s) \subseteq G_s$ lets us conclude that (j, s) belongs to T_D, so

$$S_{M,N} \subseteq \bigsqcup_D T_D.$$

Hence,

$$S_{M,N} = \bigsqcup_D T_D.$$

□

Corollary. If $(j, s) \in T_D$ and $(j', s') \in T_{D'}$, then

$$d((j, s)(j', s')) \geq \frac{\sqrt{pN}}{2M}.$$

Note on graph coverings:

We could also study some things about graph coverings. Let ℓ be a prime not dividing N and p. Let X be the ℓ-isogeny graph consisting of supersingular j-invariants as vertices and ℓ-isogenies as edges. Because of dual isogeny, we get an undirected graph.

We can observe that if ϕ connects E to E', then ϕ also connects (E, G_i) to $(E', \phi(G_i))$. Also note that since

$$\hat{\phi} \circ \phi : E[N] \to E[N]$$

has trivial kernel (since $\gcd(\deg \phi, N) = 1$), then $\{\phi(G_i)\}$ are all the order N cyclic subgroups of $E'[N]$.

$$\begin{array}{ccc} (E, G_k) & \xrightarrow{\phi} & (E', \phi(G_k)) \\ & \vdots & \\ (E, G_2) & \xrightarrow{\phi} & (E', \phi(G_2)) \\ (E, G_1) & \xrightarrow{\phi} & (E', \phi(G_1)) \\ \downarrow & & \downarrow \\ E & \xrightarrow{\phi} & E' \end{array}$$

So the covering from X_N to X_1 is definitely surjective and it's also locally bijective.

We can consider a sequence of graph coverings of the form

$$X_{n_1}, X_{n_2}, \ldots,$$

where

$$n_{i+1} = n_i p_{y_i},$$

where y_i is some indexing.

Since

$$E[n_{i+1}] = E[n_i p_{y_i}] \cong E[n_i] \times E[p_{y_i}],$$

then transition maps between different levels will be just projections, forgetting necessary coordinates.

Going to higher level and scaling by $\sqrt{p_{i+1}}$ in the limit gives dots corresponding to the isogeny islands and possibly inverse limit of these covers has some interesting structure attached to it.

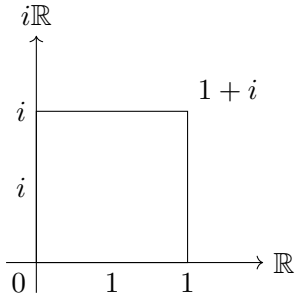
7 Proof of Theorem B

We prove that there exists a point P on $E(\mathbb{C})$ such that $\{nP\}_{n\in\mathbb{Z}}$ is dense in $E(\mathbb{C})$ which by the fact that isogenies are continuous functions from E to E will be enough to determine φ at any point of E.

Let's note that isogeny $\varphi : E \to E$ by Weierstrass equation translates to a homomorphism $f : \Pi \to \Pi$, where Π represents the fundamental parallelogram. Group operation on $E(\mathbb{C})$ is translated to group operation on Π, so if P corresponds to point $w \in \Pi$ then $\varphi(P)$ corresponds to point $f(w)$ where f is corresponding function to φ.

At this point it is easy to notice that if we find point $w \in \Pi$ for which $\{nw\}_{n\in\mathbb{Z}}$ is dense in Π, then we can by knowing w and $f(w)$ determine f at any point, hence determine φ by Weierstrass correspondence. We note that f is also continuous rather than only a homomorphism.

Without loss of generality we assume that fundamental parallelogram is of this form (otherwise make another isomorphism to this specific fundamental parallelogram):



Now let's choose "good" w. note that if we choose $w \in \Pi$ so that its x and y coordinates are irrational numbers, then $\{nw_x\}_{n\in\mathbb{Z}}$ and $\{nw_y\}_{n\in\mathbb{Z}}$ will be dense in the interval $(0, 1)$. Let's take w such that its x, y coordinates and the ratio y/x are all irrational.

Claim: $\{nw\}_{n\in\mathbb{Z}}$ is dense in Π.

Proof:

As motivation let's look at the proof of the fact that $\{nw_x\}$ is dense in $(0, 1)$.

For this case, let's think of $(0, 1)$ as a circle and note that if a and b belong to this circle, then so does their sum and difference. Divide the interval into $2N$ parts and consider first $2N + 1$ points from the sequence. Then, by the Pigeonhole Principle, we get that the distance between some 2 of these points should be less than $1/2N$. Take these points and continuously add these until they make a full turn around the circle. Because of the length being less than $1/2N$, we get that in any interval of length $1/N$ there is at least 1 point of the sequence, and this gives density of $\{nw_x\}$ in $(0, 1)$. Motivated by this argument, let's extend this idea to the fundamental parallelogram.

Remark: This idea can also be pushed to higher dimensions by slightly different argument that we use for 2 dimensions.

Remark: We can easily prove the theorem for 2 points, by letting P_1, P_2 be points corresponding to $(w_x, 0)$ and $(0, w_y)$, where $w_x, w_y \in \mathbb{R} \setminus \mathbb{Q} \cap (0, 1)$. Then it's clear how to finish this case.

If we can prove that in any small square in the parallelogram there is an integer multiple of w, we're done.

Again, we note that if a and b belong to the parallelogram, then so does their sum and difference.

Divide the parallelogram into N^2 smaller squares and consider first $N^2 + 1$ multiples of w. Again, by the Pigeonhole Principle, some 2 points will be in the same square, bounding their distance above by $\sqrt{2} \cdot 1/N$.

Take any point in the parallelogram and consider ϵ by ϵ square with the point in the center and square oriented in the same direction as fundamental parallelogram. Choose N large enough

so that some small square obtained by partitioning Π into N^2 parts is inside the ϵ by ϵ square.

Let's further partition Π into smaller squares and suppose that we can find 2 pairs of elements that are close enough (distance is less than $\sqrt{2}\epsilon$) and are not oriented in the same direction, equivalently, are not linearly dependent. Then Π contains a lattice generated by these elements and because of their size the integer multiples of these generators can't skip ϵ by ϵ square, hence there exists some integer multiple of w in ϵ by ϵ square, around arbitrarily chosen point, therefore multiples of w are dense in Π.

We now need to show that we can always find 2 pairs of points with different orientations.

Lemma: 3 points cannot lie on the same line in Π.

Proof: For the sake of contradiction suppose 3 points, $w_1, w_2, w_3 \in \Pi$ lie on the same line. These points are $k_1 w, k_2 w, k_3 w \mod$ lattice for some integers k_1, k_2, k_3.
We have that

$$k_1 w = w_1 + \tilde{l_1},$$

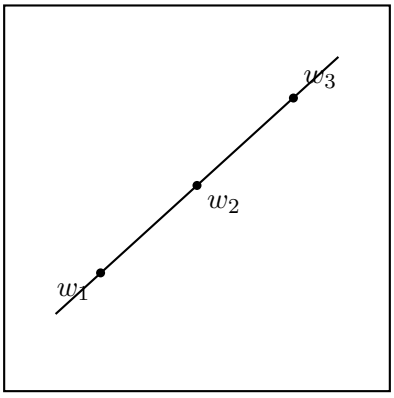
for some lattice element $\tilde{l_1}$, and analogously for $k_2 w$, $k_3 w$. So, we have that

$$w_1 = k_1 w - \tilde{l_1}$$

and

$$w_2 = k_2 w - \tilde{l}_2, w_3 = k_3 w - \tilde{l}_3,$$

Using the fact that w_1, w_2, and w_3 are on the same line in Π we get that



$$(w_2 - w_1)c = (w_1 - w_3)$$

for some scalar c, where w_i is viewed as a vector (w_{ix}, w_{iy}).

This equation is equivalent to the following:

$$(k_2 w - k_1 w + l_1)c = k_1 w - k_3 w + l_2.$$

Where l_1, l_2 are some lattice elements. Looking at this component-wise we get:

$$w_x = \frac{l_{2x} - l_{1x}}{c(k_2 - k_1) - (k_1 - k_3)},$$

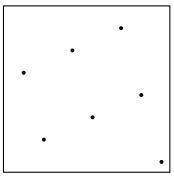
$$w_y = \frac{l_{2y} - l_{1y}}{c(k_2 - k_1) - (k_1 - k_3)}$$

and

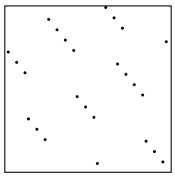
$$\frac{w_y}{w_x} = \frac{l_{2y} - l_{1y}}{l_{2x} - l_{1x}} \in \mathbb{Q},$$

which gives a contradiction to the fact that w has irrational slope. This concludes the proof.

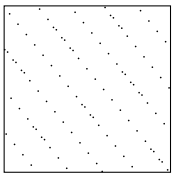
Diagrams for multiples of w, where $w = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}\right)$ is chosen for better visualization.



7 points



25 points



100 points

8 Further remarks on Theorem B

One might wish to push the idea of the previous result to the case where one defines elliptic curves over quaternion algebras. However, the non-commutative nature of quaternion algebras introduces more subtleties to consider, such as defining the addition of points in the first place, and much more.

If one defines all of these necessary ingredients and wishes to find some kind of connection between the fundamental parallelepiped of a quaternion algebra and an elliptic curve over a quaternion algebra (similarly as for $E(\mathbb{C})$ and Π, the fundamental parallelogram), one cannot really hope to use an analogous Weierstrass $\wp$ function for quaternion algebras.

Before we say anything at all, we need to define differentiation of a function defined on a quaternion algebra (if we're looking for something analogous to weierstrass correspondence between $E(\mathbb{C})$ and Π), which we can define quite naturally as

$$\varphi'(h) = \lim_{\|\alpha\| \to 0} \alpha^{-1}\big(\varphi(h+\alpha) - \varphi(h)\big).$$

By similar reasoning as in complex analysis[10], we find that φ is differentiable if and only if the E-C-R (Extended Cauchy–Riemann) conditions hold:

E-C-R:

$$\varphi = u + iv + js + kr,$$

where u, v, s, r are functions of x, y, z, t.

The E-C-R conditions are

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = \frac{\partial s}{\partial z} = \frac{\partial r}{\partial t},$$

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = -\frac{\partial r}{\partial z} = \frac{\partial s}{\partial t},$$

$$\frac{\partial s}{\partial x} = \frac{\partial r}{\partial y} = -\frac{\partial u}{\partial z} = -\frac{\partial v}{\partial t},$$

$$\frac{\partial r}{\partial x} = -\frac{\partial s}{\partial y} = \frac{\partial v}{\partial z} = -\frac{\partial u}{\partial t}.$$

After all of this, one can verify by hand that

$$\varphi(x) = x^{-2}$$

is not differentiable. Therefore, trying to establish this connection between an analogous version of the Weierstrass $\wp$ function for $\mathbb{C}$ and one for quaternion algebras should not be expected to be done using sums of a bunch of functions of the form

$$t^{-2} = \frac{1}{t^2}$$

However, one might find the E-C-R conditions handy and hopefully save some time by avoiding computations.

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